

Bistable periodic traveling waves and dynamical stabilization in an integrodifference equation with strong Allee effect and overcompensatory cycles

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1 Introduction

In this study we consider the integrodifference equation

$$u_{t+1}(x) = Qu_t(x) = \int_{-\infty}^{\infty} k(x-z) g(u_t(z)) dz \quad (1.1)$$

when the growth function g exhibits both strong Allee effect (stability at zero) and overcompensation (non-monotonicity at carrying capacity) leading to periodic cycles. We consider the following two-parameter growth function

$$g(u; a, r) = u \exp(r(1-u)(\frac{u}{a} - 1)) \quad (1.2)$$

where $r \geq 0$ is a growth rate parameter and $a \in (0, 1]$ is an Allee threshold. The dispersal kernel k is generally taken to be a symmetric probability density; in applications it is typically given by the Laplace or Gaussian distribution. We focus specifically on traveling wave solutions, which are important for modeling biological invasions and represent the limiting behavior of solutions started from a bounded domain.

This model represents a combination of two effects which have previously been studied independently:

- monotone growth models with an Allee threshold exhibit bistable traveling waves with a unique propagation speed determined by the growth function alone (and not the kernel); a recent proof was given in [LO22],
- non-monotone (overcompensatory) growth models with stable cycles but without an Allee threshold exhibit complex phenomena like period-doubling wave fronts stacked wave-trains, wherein the leading invasion wave is trailed by one or more periodic traveling waves of lesser speeds. Furthermore, the separating wave fronts leave in between them a growing plateau on which the solution takes an unstable equilibrium value in a phenomenon known as *dynamical stabilization*; the equilibrium is said to be *dynamically stabilized* by the dispersal. These were originally observed in [K92] and more recent analytical work was done in [BLL20], [BLL18].

The combinations of these effects has been studied in integrodifference equations; numerically in [SLMNS17] and simplified piecewise growth models analytically in [O17] and [NL22]. TODO: More info.

Our main contribution here is a primarily numerical investigation of traveling wave behavior in this model. Specifically, the following phenomenon which, to our knowledge, has not been described before in integrodifference models: the existence of a *periodic bistable wave* directly connecting the stable cycle to zero. We conjecture this wave exists precisely when the trailing wave connecting the stable cycle to one overtakes the leading wave connecting the carrying capacity to zero in speed. This behavior is depicted in Figure 1 for the case when the growth function has a stable 2-cycle $\{u^+, u^-\}$

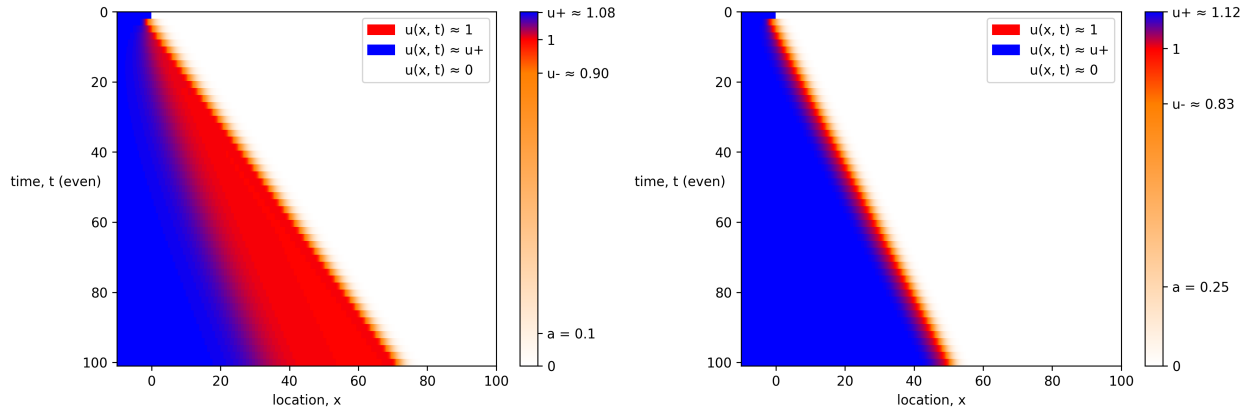


Figure 1: Two distinct outcomes from step initial data connecting $u = u^+$ to $u = 0$ plotted on even time steps $t \in \{0, 2, 4, \dots\}$. Left: with $a = 0.1$, $r = 0.23$, the solution converges to a stacked wave train consisting of a monostable 2-periodic traveling wave connecting $\{u^+, u^-\}$ to 1 followed by a monostable traveling wave connecting 1 to 0. Right: with $a = 0.25$, $r = 0.75$, the solution converges to a bistable periodic traveling wave directly connecting $\{u^+, u^-\}$ to 0.

2 Analysis

2.1 Local growth dynamics

This section contains basic facts about the growth function (1.2).

Proposition 1. The growth function (1.2) has the following properties:

- $g(0) = 0$, $g(a) = a$, and $g(1) = 1$ are the only fixed points.
- $0 < g(u) < u$ on $(0, a) \cup (1, \infty)$ and $g(u) > u$ on $(a, 1)$.

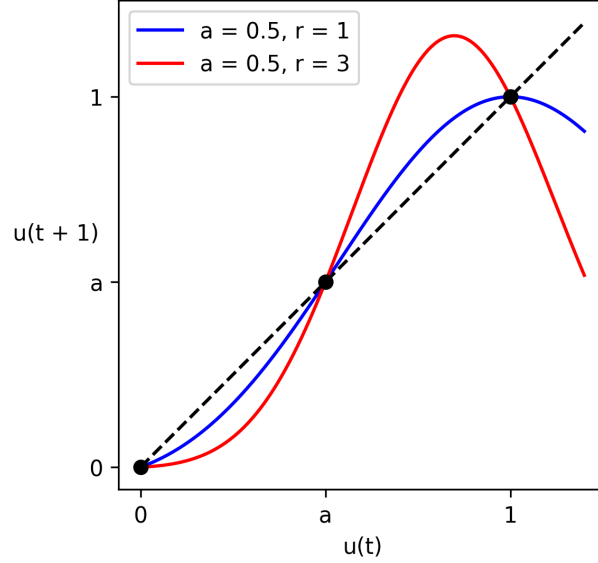


Figure 2: The growth map with a couple different choices of parameters.

iii. g is increasing on $[0, u^*]$ and decreasing on $[u^*, \infty)$ where u^* is the global maximum given by

$$u^* = \frac{1}{4} \left(a + 1 + \sqrt{(a+1)^2 + \frac{8a}{r}} \right)$$

iv. g is monotone on $[0, 1]$ iff. $r \leq \frac{a}{a-1}$.

v. $0 < g'(0) < 1$, making $u = 0$ an asymptotically stable fixed point.

vi. $g'(a) > 1$,

vii. TODO: more here

2.2 Traveling waves

In this section, we describe the traveling waves we are interested in.

A traveling wave connecting 1 to 0, which is bistable when $r < r_1$ and monostable when $r > r_1$:

$$Qw_1(x) = w_1(x - c_1), \quad w_1(-\infty) = 1, \quad w_1(+\infty) = 0 \quad (\text{TW})$$

When $\{u^+, u^-\}$ form a stable 2-cycle, a monostable 2-periodic wave connecting $u^\pm$ to 1:

$$Q^2 w_2(x) = w_2(x - 2c_2), \quad w_2(-\infty) = u^+, \quad w_2(+\infty) = 1 \quad (\text{MPTW})$$

When $\{u^+, u^-\}$ form a stable 2-cycle, a bistable 2-periodic wave connecting $u^\pm$ directly to 0:

$$Q^2 w(x) = w(x - 2c), \quad w(-\infty) = u^+, \quad w(+\infty) = 0 \quad (\text{BPTW})$$

2.3 Initial value problems

We are also interested in the following initial value problems, both when $r_1 < r < r_2$:

With step initial data:

$$\begin{aligned} u_{t+1} &= Qu_t \\ u_0(x) &= \begin{cases} u^+ & x \leq 0 \\ 0 & x > 0 \end{cases} \end{aligned} \quad (\text{IVP})$$

and with compact initial data on an interval of radius $R > 0$:

$$\begin{aligned} u_{t+1} &= Qu_t \\ u_0(x) &= \begin{cases} u^+ & |x| \leq R \\ 0 & |x| > R \end{cases} \end{aligned} \quad (\text{IVP})$$

2.4 Case 1: monotone growth

When the growth function is monotone, we can apply the results of [LO22] to obtain the existence of a traveling wave with a unique propagation speed. The following is an immediate application of Theorem 3.1 in [LO22]:

Proposition 2. Suppose $r \leq r_m$. Then there exists a unique c_1^* such that there exists a nonincreasing traveling wave $w_1 \in C(\mathbb{R}, [0, 1])$ such that

$$Qw_1(x) = w_1(x - c_1^*), \quad w_1(-\infty) = 1, \quad w_1(+\infty) = 0$$

2.5 Case 2: nonmonotone growth, stable carrying capacity

2.6 Case 3: nonmonotone growth, stable 2-cycles

When $r_1 < r < r_m^1$ we can apply the results of Theorem 3.1 in [BLL18]:

Proposition 3. Suppose $r_1 < r \leq r_m^1$ (so that g^2 is monotone on $[u^-, u^+]$). Then there exists a *spreading speed* c_2^* for the double-iterate operator Q^2 from 1 to u^+ . This means:

1. for any initial data $u_0 \in C(\mathbb{R}, [1, u^+])$ with $u_0(x) = 1$ outside a compact set, then

$$\lim_{t \rightarrow \infty} \sup_{|x| \geq ct} Q^{2t} u_0(x) = 1, \quad \forall c > c_2^*$$

2. for any initial data $u_0 \neq 1$,

$$\lim_{t \rightarrow \infty} \inf_{|x| \leq ct} Q^{2t} u_0(x) = u^+, \quad \forall c \in (0, c_2^*)$$

We also get the existence of traveling waves from Theorem 3.2 in [BLL18]:

Proposition 4. If $r_1 < r \leq r_m^1$, then with c_2^* given in the previous theorem, there exists a monotone traveling wave solution $u_{2t}(x) = w_2(x - 2c_2^*t)$ such that $w_2(-\infty) = u^+$ and $w_2(+\infty) = 1$.

TODO: more here?

2.7 Else: nonmonotone growth, higher cycles and chaos

3 Numerical simulations and conjectures

3.1 The initial value problem with step initial data connecting u^+ to 0

In this section we consider the initial value problem

$$\begin{aligned} u_0(x) &= \begin{cases} u^+ & x \leq 0 \\ 0 & x > 0 \end{cases} \\ u_{t+1}(x) &= Qu_t(x) \end{aligned} \quad (3.1)$$

when the growth function has a stable 2-cycle, i.e. when $r_1(a) < r < r_2(a)$. Our basic observation, as depicted in [Figure 1](#), is that depending on the choice of parameters, the solution converges either to a 2-periodic *bistable wave* directly connecting u^+ to 0, or a pair of *stacked waves* moving at different velocities connecting u^+ to 1 and 1 to 0 respectively.

We make this prediction precise in the following conjecture:

Conjecture 5. The solution $u_t(x)$ to the initial value problem (3.1) falls in one of two cases depending on the growth parameters (a, r) :

- Case 1 (stacked wave): the solution converges uniformly to a superposition of traveling waves, that is

$$\sup_{x \in \mathbb{R}} \|u_{2t}(x) - w_1(x - 2c_1t) - w_2(x - 2c_2t) + 1\|_\infty \rightarrow 0$$

where w_1 is a solution to (??) and w_2 is a solution to (??).

- Case 2 (bistable wave): the solution converges uniformly to a bistable traveling wave, that is

$$\sup_{x \in \mathbb{R}} \|u_{2t}(x) - w(x - 2c_2t)\|_\infty \rightarrow 0$$

where w is a solution to (??)

Remark 6. Notice that our characterization of stacked waves only requires uniform convergence on compact sets.

Based on this conjecture, we can design a numerical experiment to estimate the wave speeds in either case. Attach to the solution the following sequence of front-tracking points, for a given $\epsilon > 0$:

$$\alpha_t = \sup\{x \mid |u_t(x) - g^t(u^+)| \geq \epsilon\}, \quad \beta_t = \inf\{x \mid |u_t(x)| \geq \epsilon\}$$

and the wave speed estimates

$$c_2(t) = \frac{1}{2}(\alpha_{t+2} - \alpha_t), \quad c_1(t) = \beta_{t+1} - \beta_t$$

$$c_t^\pm(\ell, \epsilon) = (\xi_{t+1}^\pm - \xi_t^\pm)(\ell, \epsilon)$$

If the solution falls into case 1 and converges to a bistable 2-periodic wave w_t with velocity c , then

$$\lim_{t \rightarrow \infty} c_t^+(u^+, \epsilon) = \lim_{t \rightarrow \infty} c_t^\pm(1, \epsilon) = \lim_{t \rightarrow \infty} c_t^-(0, \epsilon) = c$$

(as long as $\epsilon < \min\{u^+ - 1, 1 - u^-, u^-\}$).

On the other hand, if u_t converges to a stacked traveling wave consisting of w_t^1 and w_t^2 with velocities c_1 and c_2 respectively, and assuming w_t^2 is uniformly bounded away from zero and w_t^1 is uniformly bounded away from u^+ , that is,

$$\exists \delta, \forall t, \quad \inf w_t^2 \geq \delta > 0, \quad \sup w_t^1 \leq u^+ - \delta < u^+$$

then

$$\lim_{t \rightarrow \infty} c_t^+(u^+, \epsilon) = \lim_{t \rightarrow \infty} c_t^-(1, \epsilon) = c_2$$

and

$$\lim_{t \rightarrow \infty} c_t^+(1, \epsilon) = \lim_{t \rightarrow \infty} c_t^-(0, \epsilon) = c_1$$

(For a proof of this claim refer to the appendix. TODO)

Our next plot is based on the following observation:

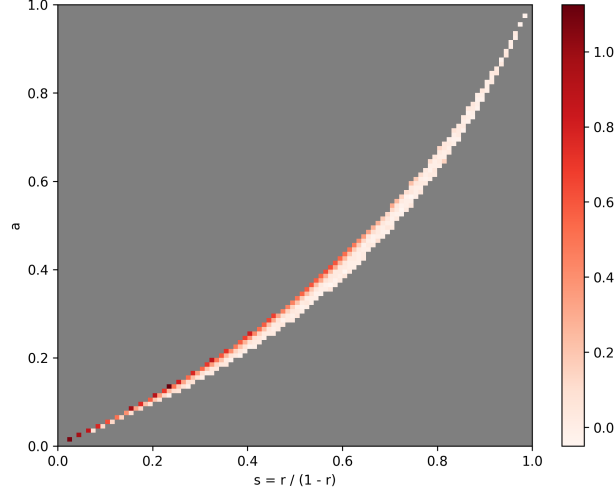
$$\lim_{t \rightarrow \infty} c_t^+(u^+, \epsilon) - c_t^-(0, \epsilon) = \begin{cases} 0 & \text{in case 1} \\ c_1 - c_2 & \text{in case 2} \end{cases}$$

Figure ?? shows the result of approximating the above limit for a range of (a, r) values. We observe that both behaviors occur on a nontrivial region of parameter space.

A couple more conjectures to organize: (TODO)

Conjecture 7. Suppose

Conjecture 8. For sufficiently small $a \in (0, 1)$ there exists a threshold $r_{thres}(a) \in (r_1(a), r_2(a))$ below which case 2 occurs, and above which case 1 occurs.



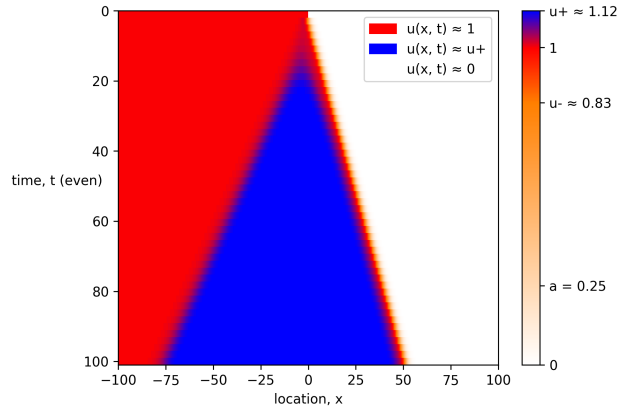
3.2 Comparing wavespeeds via continuation method

In the previous section, we conjectured two possible outcomes of the initial value problem (3.1) when $r_1(a) < r < r_2(a)$. In the second case when two traveling waves emerged, our numerical algorithm allowed us to estimate both wave speeds. In the first case however, when a bistable wave emerges, we gain no information about the existence of waves connecting $u^\pm$ to 1 or 1 to 0. For example, consider Figure ??, which depicts the initial value problem

$$u_0(x) = \begin{cases} 1 & x \leq 0 \\ 0 & x > 0 \end{cases} \quad (3.2)$$

$$u_{t+1}(x) = Qu_t(x)$$

with growth parameters $a = 0.25$, $r = 0.75$ (for which a bistable wave appears in problem (3.1)). Rather than converge to a traveling wave connecting 1 to 0, the solution converges to a pair of outgoing waves, connecting $u^\pm$ to 1 on the left and $u^\pm$ to 0 on the right.



Thus, if there is a wave connecting 1 to 0, we are unable to observe it from the initial value problem. Instead, we consider a continuation method as follows which slowly increases the growth rate parameter r from the starting value $r_m(a)$, at which the growth function is monotone and a $(1, 0)$ wave exists by theorem, and slowly increases to the value we are interested in.

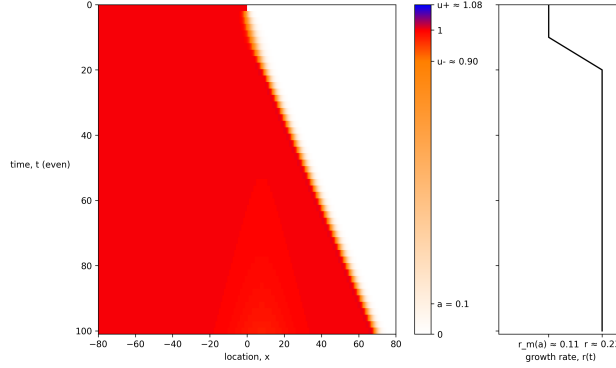
Specifically we set up the following problem for a given (r, a) such that $r_1(a) < r < r_2(a)$:

$$\begin{aligned}
u_0(x) &= \begin{cases} 1 & x \leq 0 \\ 0 & x > 0 \end{cases} \\
u_{t+1}(x) &= Q_t u_t(x) \\
Q_t[u] &= k * (g_t \circ u) \\
g_t(u) &= g(u; a, r_t) \\
r_t &= \begin{cases} r_m(a) & 0 \leq t < T_0 \\ r_m(a) + \frac{t-T_0}{T_1-T_0}(r - r_m(a)) & T_0 \leq t < T_1 \\ r & T_1 \leq t \end{cases}
\end{aligned} \tag{3.3}$$

where T_0 and T_1 are scheduling parameters. To justify this setup, we make the following conjecture:

Conjecture 9. If there exists a traveling wave $w(x)$ connecting 1 to 0, then the solution $u_t(x)$ to (3.3) will converge to it for sufficiently large T_0, T_1 .

TODO: I don't even think this is true.



3.3 Detecting dynamical stabilization

3.4 Main conjecture

Our main conjecture is the following:

Conjecture 10.

4 Appendix

4.1 Traveling waves and stacked waves

In this section we comment on the analytical interpretation of so-called ‘wave-train’ solutions, aka stacked traveling waves or terraces.

Recall some basic facts about traveling waves:

1. A solution $u_t(x)$ is called a traveling wave with (rightward) velocity c if it satisfies

$$u_{t+1}(x) = u_t(x - c)$$

Any traveling wave has the form

$$u_t(x) = w(x - ct)$$

where w is a solution to the equation

$$Qw(x) = w(x - c)$$

We may call w a traveling wave profile with velocity c .

2. A traveling front is a traveling waves with limits at $\pm\infty$, which are necessarily both fixed points of g , that is

$$g(u_t(\pm\infty)) = u_t(\pm\infty)$$

3. A solution $u_t(x)$ is called a periodic traveling wave with period p and mean velocity c if it satisfies

$$u_{t+p}(x) = u_t(x - cp)$$

Any periodic traveling wave has the form

$$u_t(x) = w_t(x - pc)$$

where w solves the equation

$$Q^p w_t(x) = w_t(x - pc)$$

and is periodic in t ,

$$w_{t+p} = w_t$$

We can say that w_t is a periodic traveling wave profile.

4. A p -periodic traveling front is a p -periodic traveling wave with limits at $\pm\infty$, which are necessarily both p -periodic points of g , that is,

$$g^p(u_t(\pm\infty)) = u_t(\pm\infty)$$

5. Alternatively we can just define a periodic traveling wave of Q to be a traveling wave of Q^p .
6. We say that a solution $u_t(x)$ converges uniformly to a traveling wave if

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w(x - c)| = 0$$

where w_t is a traveling wave profile with velocity c .

7. We say that a solution $u_t(x)$ converges uniformly to a periodic traveling wave if

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w_t(x - pc)| = 0$$

where w_t is a periodic traveling wave profile with mean velocity c .

It's not immediately clear how to adopt the above definitions to convergence to a stacked wave.

Proposition 11. Suppose $u_t(x)$ converges to a traveling wave, that is

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w(x - c^*t)| = 0$$

for some traveling wave profile w with velocity c^* , and suppose w has limits at $\pm\infty$ Then:

1. If $c < c^*$ then $u_t(x + ct) \rightarrow w(-\infty)$ uniformly on compact sets.
2. If $c = c^*$ then $u_t(x + ct) \rightarrow w$ uniformly.
3. If $c > c^*$ then $u_t(x + ct) \rightarrow w(+\infty)$ uniformly on compact sets.

Proof. 1. Suppose $c < c^*$ and let K be a compact set. By the triangle inequality, for all x and t ,

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w(-\infty)| &\leq \sup_{x \in K} |u_t(x + ct) - w(x - (c^* - c)t)| \\ &\quad + \sup_{x \in K} |w(x - (c^* - c)t) - w(-\infty)| \end{aligned}$$

It suffices to show both terms on the right vanish as $t \rightarrow \infty$.

For the first term we have

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w(x - (c^* - c)t)| &= \sup_{x \in (K + ct)} |u_t(x) - w(x - c^*t)| \\ &\leq \sup_{x \in \mathbb{R}} |u_t(x) - w(x - c^*t)| \end{aligned}$$

which vanishes by assumption that u_t converges to a traveling wave.

For the second part, $\epsilon > 0$. We need to find t_0 such that

$$\sup_{x \in K} |w(x - (c^* - c)t) - w(-\infty)| < \epsilon, \quad \forall t \geq t_0$$

Let x_0 be such that $|w(x) - w(-\infty)| < \epsilon$ for all $x \leq x_0$. It suffices to choose t_0 such that

$$x - (c^* - c)t \leq x_0, \quad \forall x \in K, \quad \forall t \geq t_0$$

For this it suffices to have

$$\sup K - (c^* - c)t \leq x_0, \quad \forall t \geq t_0$$

Solving this inequality shows that it suffices to take any $t_0 \geq \frac{x_0 - \sup K}{c^* - c}$.

2. If $c = c^*$ then simply apply translation inside the supremum:

$$\lim_{t \rightarrow \infty} \sup_x |u_t(x + c^*t) - w(x)| = \lim_{t \rightarrow \infty} \sup_x |u_t(x) - w(x - c^*t)| = 0$$

3. The third case mirrors the first case. □

The next theorem generalizes this to periodic traveling waves.

Remark 12. The next proposition assumes w_t has limits at $\pm\infty$ *uniformly in t* . This means:

$$\forall \epsilon > 0, \exists x_0, \forall t, \forall x \leq x_0, |w_t(x) - w_t(-\infty)| < \epsilon$$

(and likewise at $+\infty$).

Proposition 13. Suppose $u_t(x)$ converges to a periodic traveling wave, that is

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w_t(x - c^*t)| = 0$$

for some periodic traveling wave profile w_t with mean velocity c^* , and suppose w_t has limits at $\pm\infty$ *uniformly in t* . Then:

1. If $c < c^*$ then $u_t(x + ct) \rightarrow w_t(-\infty)$ uniformly on compact sets.
2. If $c = c^*$ then $u_t(x + ct) \rightarrow w_t$ uniformly.
3. If $c > c^*$ then $u_t(x + ct) \rightarrow w_t(+\infty)$ uniformly on compact sets.

Or if we want to be more precise:

1. If $c < c^*$ then $u_t(x + ct) - w_t(-\infty) \rightarrow 0$ uniformly on compact sets.
2. If $c = c^*$ then $u_t(x + ct) - w_t(x) \rightarrow 0$ uniformly.
3. If $c > c^*$ then $u_t(x + ct) - w_t(+\infty) \rightarrow 0$ uniformly on compact sets.

Proof. 1. Suppose $c < c^*$ and let K be a compact set. By the triangle inequality, for all x and t ,

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w_t(-\infty)| &\leq \sup_{x \in K} |u_t(x + ct) - w_t(x - (c^* - c)t)| \\ &\quad + \sup_{x \in K} |w_t(x - (c^* - c)t) - w_t(-\infty)| \end{aligned}$$

It suffices to show both terms on the right vanish as $t \rightarrow \infty$.

For the first term we have

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w_t(x - (c^* - c)t)| &= \sup_{x \in (K + ct)} |u_t(x) - w_t(x - c^*t)| \\ &\leq \sup_{x \in \mathbb{R}} |u_t(x) - w_t(x - c^*t)| \end{aligned}$$

which vanishes as $t \rightarrow \infty$ by assumption that u_t converges to a traveling wave.

For the second part, $\epsilon > 0$. We need to find t_0 such that

$$\sup_{x \in K} |w_t(x - (c^* - c)t) - w_t(-\infty)| < \epsilon, \quad \forall t \geq t_0$$

Let x_0 be such that $|w_t(x) - w_t(-\infty)| < \epsilon$ for all $x \leq x_0$ and for all t . It suffices to choose t_0 such that

$$x - (c^* - c)t \leq x_0, \quad \forall x \in K, \quad \forall t \geq t_0$$

For this it suffices to have

$$\sup K - (c^* - c)t \leq x_0, \quad \forall t \geq t_0$$

Solving this inequality shows that it suffices to take any $t_0 \geq \frac{x_0 - \sup K}{c^* - c}$.

2. If $c = c^*$ then simply apply translation inside the supremum:

$$\lim_{t \rightarrow \infty} \sup_x |u_t(x + c^*t) - w_t(x)| = \lim_{t \rightarrow \infty} \sup_x |u_t(x) - w_t(x - c^*t)| = 0$$

3. The third case mirrors the first case.

□

4.2 Generalized defn